



Chapter 7

Object detection and segmentation of trees using Text SAM in ArcGIS Online

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Overview

In this chapter, you will learn how to use GIST tools and methodologies to analyze urban greenery by integrating aerial imagery and deep learning techniques. You will focus on applying the Segment Anything Model (SAM) for tree and building segmentation and explore how these technologies address the time and cost limitations of traditional manual monitoring and planning, particularly in large-scale and complex urban environments. Specifically, you will learn a nonprogramming approach to identifying and segmenting trees and buildings from aerial imagery using Text SAM and ArcGIS Online without relying on Python scripting. This process-oriented analytic method provides decision-makers and urban planners with precise, efficient greening data and strategic insights, enhancing their ability to implement scientifically optimized strategies. By offering data faster and more cost-effectively than conventional methods, this approach has the potential to transform urban planning, enabling more informed and data-driven decisions for sustainable city development.

Note: Text SAM is an open-source sample model that can be prompted using free-form text prompts to extract features of various kinds. This is achieved by using the Grounding DINO model and SAM.

Learning objectives

- Convert aerial TIF images into image layers using ArcGIS Online.
- Apply deep learning tools for accurate identification and segmentation of trees without coding, using the Text SAM model and Map Viewer in ArcGIS Online.

- Use aerial imagery for exploratory data visualization to assess the current state of urban greenery.
- Evaluate discrepancies between current urban conditions and planning objectives to propose data-driven optimization strategies for sustainable city development.

Technical components

- Software:
 - ArcGIS Image for ArcGIS Online
 - ArcGIS Online
 - ArcGIS Pro 3.4
- Datasets:
 - Aerial map of Estella Public School in New South Wales, Australia, available for download at links.esri.com/OpenAerialMap
 - Text SAM available for download at www.arcgis.com/home/item.html?id=8df3bf4167bc4c7b967f677f8b362ec3

Prerequisites

- Beginner to intermediate proficiency with ArcGIS Online.
- Basic knowledge of or experience with ArcGIS Online Map Viewer.
- No prior knowledge or experience with the Python programming language, machine learning, or deep learning required.
- Technical skill level: Introductory GIS, introductory deep learning.
- Time to complete chapter: one hour without intelligence brief, two to three hours with intelligence brief.

Background and situational context

With the rapid acceleration of global urbanization, urban greenery has become a central issue in environmental governance and sustainable community development. Well-planned urban greenery plays a crucial role in regulating microclimates, improving quality of life, mitigating the urban heat island effect, enhancing air quality, and strengthening the stability of urban ecosystems. However, owing to the complexity of urban structures and the diversity of green spaces, traditional manual monitoring and planning methods often face challenges related to efficiency, cost, and precision management.

In recent years, advancements in remote sensing technology and deep learning algorithms have introduced groundbreaking opportunities for urban green space planning, monitoring, and optimization. The use of aerial imagery allows real-time capture of urban greenery, and AI-powered object detection and segmentation techniques enable precise analysis. Compared with traditional methods, deep learning models can automatically identify and segment objects, such as trees, buildings, and roads,

with high accuracy. This not only provides comprehensive data on green space types, spatial distribution, and coverage but also helps identify areas requiring improved greenery, offering data-driven recommendations for urban planners.

To make these advanced AI techniques accessible to nonprogrammers, we have chosen to use ArcGIS Online, enabling users to perform object detection, recognition, and segmentation without the need for programming. Additionally, this approach supports the integration of ArcGIS Notebooks and ArcGIS API for Python, using the pretrained SAM model for tree identification and segmentation. Without using complex coding, users can complete data processing and analysis through an intuitive visual interface, allowing urban planners and city managers to harness the power of AI and GIS for more precise and efficient urban greenery management.

In this chapter, we provide a detailed guide on how to use ArcGIS Online, Text SAM, and other tools to perform automated tree and building detection and segmentation from aerial imagery in a no-code environment, ultimately supporting sustainable urban development and intelligent planning.

Exercise workflow 7

1 Load the imagery and add metadata

Note: This step requires the license for ArcGIS Image for ArcGIS Online. You can confirm that you have this license by clicking your profile icon in the top right > My settings > Licenses, and search for ArcGIS Image for ArcGIS Online.

1. In a browser, navigate to links.esri.com/OpenAerialMap and download the image of Estella Public School in New South Wales, Australia, to your computer. Rename the file `Estella_Public_School.tif`.
2. In a browser, navigate to www.arcgis.com and sign in using your ArcGIS organization account.
3. On the ribbon, click the **Content** tab. On the top left, click **New Item**.
4. In the **New item** window, select **Imagery layer**.

The **Create imagery layers window** opens.

5. Check the box for **Dynamic Imagery Layer**. Uncheck the box for **Tiled Imagery Layer**. Click **Next**.

To prepare for your subsequent Text SAM tasks, it is best to select **Dynamic Imagery Layer**. Tiled imagery layers are suitable for fast display and access to static images. Dynamic imagery layers provide more powerful processing and are suitable for scenarios that require flexible analysis.

6. Select **One Image** and click **Next**.

Choosing **One Image** simplifies data management, improves computing efficiency, and is compatible with deep learning and GIS analysis. When performing aerial image segmentation in ArcGIS Online, using a single dynamic image layer is a reasonable choice.

7. Select the **Estella_Public_School.tif** image from your computer and upload it.
8. Under **Set Item Details**, name the imagery **Estella_Public_School_[your initials]**, add **aerial imagery**, **deep learning**, and **SAM** as tags, and type **A high-resolution aerial image of the Estella Public School in New South Wales, Australia** as the description. Click **Create**.

Tip: This step may take several minutes to complete.

If someone else in your ArcGIS Online organization has already completed this step, it is likely that there will be a dataset with that name (Estella_Public_School) already. To save a distinct copy of it, it is recommended that you include your initials in the name.

Title

Estella_Public_School_YI

Tags

aerial imagery

deep learning

SAM

Add tags

Summary

Add summary

Save in folder

SecurityFirst

9. When the **Dynamic Imagery Layer** is created, open it in **Map Viewer** to confirm that it rendered correctly. After confirming, close the map without saving it.



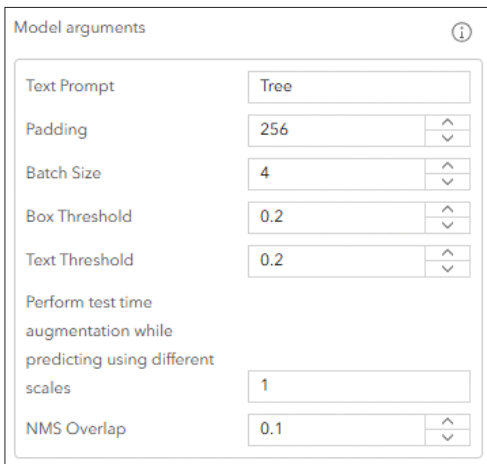
2 Download and run the Text SAM model

Next, you will download the Text SAM model.

1. In a browser, go to ArcGIS Living Atlas of the World (livingatlas.com), search for **TextSAM**, and download the model.
2. Return to your ArcGIS Online home page and confirm that you are still signed in. On the **Content** page, click **New Item** and upload the Text SAM model that you just downloaded.

Note: The size of this package exceeds 1 GB.

3. On the ribbon, click **Map** to access Map Viewer. Click **Add layer**. From **My Content**, click the dynamic imagery layer with your initials.
4. In the **Settings** pane, click **Analysis**. In the search bar, type **detect objects using deep learning** to find this deep learning tool. Alternatively, you can navigate to the **Use Deep Learning** toolbox to locate the **Detect Objects Using Deep Learning** tool.
5. Open the **Detect Objects Using Deep Learning** tool and add the following settings:
 - **Input imagery or feature layer:** Estella_Public_School_[your initials]
 - **Processing mode:** Process as mosaicked image
 - **Model settings:** Select **Text SAM** model from **My Content**
6. When the **Text SAM** model loads, add the following model arguments, and click **Run**.
 - **Text Prompt:** Tree
 - **Padding:** 256
 - **Batch Size:** 4
 - **Box threshold:** 0.2
 - **Text Threshold:** 0.2
 - **Perform test time augmentation while predicting using different scales:** 1
 - **NMS Overlap:** 0.1
 - **Output Class Label Field Name:** ClassLabel
 - **Output Name:** Estella_Segmented_Trees_[your initials]



The screenshot shows the 'Model arguments' dialog box for the Text SAM model. It contains the following settings:

Argument	Value
Text Prompt	Tree
Padding	256
Batch Size	4
Box Threshold	0.2
Text Threshold	0.2
Perform test time augmentation while predicting using different scales	1
NMS Overlap	0.1

Table 1 provides descriptions of each of these model arguments.

Table 1. Model arguments

Argument	Value	Description
Text Prompt	Tree	The text input to the Text SAM model, instructing the model to detect objects related to “tree.”
Padding	256	When processing images, add 256 pixels of padding to the image to confirm that the target object is not lost owing to boundary cropping, which improves the detection effect.
Batch Size	4	The batch size of images processed at a time, set to “4.” This means that the model will process four images at the same time in one iteration to improve computational efficiency.
Box Threshold	0.2	Target box (bounding box) confidence threshold; objects below 0.2 will be filtered to avoid detecting less-reliable objects.
Text Threshold	0.2	Text-matching confidence threshold, used to control the matching degree of text prompts (such as “tree”) and detected objects; detection results below 0.2 will be ignored.
Perform test time augmentation while predicting using different scales	1	Enable test time augmentation (TTA) to use different scales when predicting to improve the success of detection and reduce the impact of scale changes on the results.
NMS Overlap	0.1	Non-maximum suppression (NMS) overlap threshold, used to remove overlapping detection boxes. Setting it to 0.1 means that if the overlap between two target boxes exceeds 10 percent, only the one with the highest confidence will be retained to reduce redundant detection results.
Output Class Label Field Name	ClassLabel	
Output Name	Estella_Segmented_Trees_[your initials]	

7. When the model has run, add the new dataset, **Estella_Segmented_Trees_[your initials]**, to your map and review it. Take note of the segmented trees.



Interpretation of analysis

After completing this exercise, answer the following questions in detail. The answers to these questions, as well as the analysis results from the exercise workflow, will inform the intelligence brief.

1. What factors might affect the accuracy of Text SAM's object detection in ArcGIS Online, and how can you validate the results?
2. How do the quality and type of imagery affect the effectiveness of Text SAM, and what imagery preprocessing steps might improve performance?
3. In what ways might biases in training data or model limitations affect decision-making in geospatial analysis using Text SAM?
4. How might you optimize the workflow for large-scale analyses while maintaining efficiency in ArcGIS Online?
5. How might Text SAM be integrated with other geospatial tools or datasets to enhance real-world applications, such as disaster response or recovery, and other scenarios?

Intelligence brief

Based on the analysis from the exercise workflow, produce a three- to five-page intelligence brief by responding to the following question prompts. Incorporate your geospatial outputs from the exercise to enrich and illustrate your conclusions.

- **BLUF:** List three to five primary observations from your analysis in bulleted-list format that summarize findings of this work.
- **Top-level analysis:** Report the primary data points supporting your BLUF.
- **Secondary analysis:** Provide the data sources and deeper analytic products from this analysis.
- **Impact:** Describe the impacts of your findings on the department or organization.
- **Confidence:** Report the level of confidence of your assessment based on available data, accuracy, and so on.
- **Bias:** Explain any background or information you have that could have affected your analysis.
- **Conclusions:** Determine what type of analytic conclusions there may be and how the data requirements, analysis, and brief writing may change based on what analytic framework you are following (descriptive, predictive, prescriptive).

Applying and extending your skills

SAM is a conceptual representation of a model that combines vision (image segmentation) with language (text cues) to segment images based on given cues. The idea is compelling because it allows the use of natural language descriptions to segment objects or features in images without extensive training on specific datasets. The actual accuracy of a SAM model (or any model with similar capabilities) depends on several factors:

- **Training data:** The variety and quality of the data on which the model is trained plays a crucial role. If a model has seen a wide variety of scenes, objects, and contexts during training, it is more likely to perform well on different test data.
- **Prompt specificity:** The specificity or vagueness of the text prompt affects the performance of the model. For example, “trees with leaves in autumn” might produce more accurate results than “trees” in general if the model understands this distinction.
- **Image complexity:** Images with clear, unique features are easier to segment than cluttered images with overlapping objects and different patterns.
- **Postprocessing and thresholding:** The way that the results are postprocessed (for example, setting confidence level thresholds) can affect the accuracy of perception. Sometimes, adjusting these parameters, depending on the context, can lead to better or worse results.

SAM is poised for significant advancement as a useful visual model in remote sensing, with extensive research focused on its enhancement. A notable development is Efficient SAM, a segmentation model that uses SAM’s mask image pretraining along a cross-attention decoder. This innovation enables Efficient SAM to deliver high performance and speed, even when deployed on smaller models.

Understanding and applying SAM technologies is increasingly crucial, given its potential to transform the landscape of remote sensing and geospatial analysis.

Additional resources

Esri Academy (www.esri.com/training) offers a variety of instructor-led and self-paced training courses about ArcGIS and its applications in different fields, including mineral exploration and environmental monitoring.

In addition, you can search *ArcGIS Blog* (esri.com/arcgis-blog/overview) for blog articles on Text SAM and other topics, including the following:

- “Multiresolution Object Detection with Text SAM” (May 16, 2024)
- “Text SAM: Extracting GIS Features Using Text Prompts” (May 14, 2024)
- “Advancing Image Segmentation with SAM: Segment Anything Model” (May 14, 2024)

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About the authors

Yifan Yang is currently pursuing a PhD in geography at Texas A&M University. He graduated from the University of Southern California with a master of science degree in spatial data science in May 2024. His research interests include spatial data science and geographic artificial intelligence. He was recently awarded a lifetime membership in the Gamma Theta Upsilon Nu Theta Chapter of the International Geographic Honor Society. He also won the ArcGIS StoryMaps Competition Award at the 2024 Los Angeles Geospatial Summit for “Best Analytical Methods Using Maps.”

Dominic Borrelli currently works as an ArcGIS Enterprise consultant at Esri Australia, where he provides technical support, consulting, and training services to Esri clients based in Australia. He graduated from the University of Southern California in 2024, completing both a bachelor of science in environmental studies and a master of science in spatial data science in May and August, respectively. Dominic has several years of experience in the geospatial industry, with a wide breadth of knowledge including ArcGIS Enterprise, Python programming for GIS, and deploying deep learning for aerial imagery analysis.



Chapter 8

Estimating populations vulnerable to flood risk in Florida between 2001 and 2019

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Overview

In this chapter, you will learn how to use GIST tools and methods to identify urban development in flood zones, analyze its change during the past 20 years, and assess the population living in flood zones in Florida in the context of HSGI. The workflows and the final deliverable will provide information to enhance public awareness of flood risks and improve future disaster preparation.

Learning objectives

- Work with publicly available land-use data and conduct spatial and temporal analysis on raster datasets using Python.
- Compare the past (2001) and near-present (2019) urban flood exposure in Florida.
- Perform exploratory data visualization using interactive maps to identify changes in urban flood exposure in Florida using ArcGIS Notebooks and ArcGIS Online.
- Estimate populations living in flood-prone areas using dasymetric mapping and visualize the estimation and its ratio to the total population.